If astronomical observations did not strongly suggest that the cores of galaxies are inhabited by gigantic black holes, astrophysicists even today, in the 1990s, would probably not predict it. However, since the observations *do* suggest gigantic holes, astrophysicists easily accommodate themselves to the suggestion. This is indicative of our poor understanding of what really goes on in the cores of galaxies.

What of the future? Need we worry that the gigantic hole in our Milky Way galaxy might swallow the Earth? A few numbers set one's mind at ease. Our galaxy's central hole (if it indeed exists) weighs about 3 million times what the Sun weighs, and thus has a circumference of about 50 million kilometers, or 200 light-seconds—about onetenth the circumference of the Earth's orbit around the Sun. This is tiny by comparison with the size of the galaxy itself. Our Earth, along with the Sun, is orbiting around the galaxy's center on an orbit with a circumference of 200,000 light-years—about 30 billion times larger than the circumference of the hole. If the hole were ultimately to swallow most of the mass of the galaxy, its circumference would expand only to about 1 light-year, still 200,000 times smaller than the circumference of our orbit.

Of course, in the roughly 10^{18} years (100 million times the Universe's present age) that it will require for our central hole to swallow a large fraction of the mass of our galaxy, the orbit of the Earth and Sun will change substantially. It is not possible to predict the details of those changes, since we do not know well enough the locations and motions of all the other stars that the Sun and Earth may encounter during 10^{18} years. Thus, we cannot predict whether the Earth and Sun will wind up, ultimately, inside the galaxy's central hole, or will be flung out of the galaxy. However, we can be confident that, if the Earth ultimately gets swallowed, its demise is roughly 10^{18} years in the future—so far off that many other catastrophes will almost certainly befall the Earth and humanity in the meantime.

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1. By *light,* I always mean in this book the type of electromagnetic waves that the human eye can see; that is, optical radiation.

2. See Figure 7.3. Ginzburg is best known not for these discoveries, but for yet another: his development, with Lev Landau, of the "Ginzburg–Landau theory" of superconductivity (that is, an explanation for how it is that some metals, when made very cold, lose all their resistance to the flow